

Civic Complaint & Grievance App

AI-Powered Citizen Reporting Platform

Nepal · Web + Mobile · Production App · English

General Public · Public Complaint Map · Account-Based · Email Dispatch

Confirmed Scope

Target audience	General public — all ages, mixed digital literacy. UI must be simple, large tap targets, minimal steps.
Complaint visibility	Public — all submitted complaints appear on a shared map. Other citizens can view and upvote active issues.
Language at launch	English only. Nepali support planned for a future phase.
Dispatch channel	Email only at launch. Complaints sent as formatted emails to the relevant government authority inbox.
Authentication	Account required. Citizens must sign up before submitting. Enables complaint history and accountability.
Product type	Production app — built to scale, not a prototype. Architecture and decisions must reflect this.

Tracking / status

Not in scope currently — no government APIs available. Complaint history stored locally in the app.

What the App Does

A production-grade web and mobile application where registered citizens report civic problems by photographing the issue and adding a short description. AI reads the image and text, classifies the problem, identifies the responsible government authority, drafts a formal complaint letter, and sends it by email on the citizen's behalf. All complaints are plotted on a public map so the community can see active issues and upvote those affecting them. Citizens can view their own submission history from their account. Complaint tracking and authority response integration are deferred to a future phase once government APIs are available.

Problem Statement

No clear channel

Citizens don't know which office handles which issue. Complaints get misdirected or ignored.

Process burden

Filing a formal complaint requires navigating bureaucratic procedures most people are unfamiliar with.

No paper trail

Verbal and informal complaints are easily lost with no documented record for accountability.

Low awareness

Most civic problems go unreported simply because citizens don't know a formal process exists.

No community view

There is no shared visibility into which problems exist where, making it hard to prioritise at scale.

How It Works — 4-Step Complaint Journey

Capture**1**

Signed-in citizen opens the app, takes a photo of the issue, and fills in a short description. The app auto-fills location from GPS; citizen can correct the address if needed.

AI Reads

2

Claude Vision API analyses the photo to identify what is visible. It combines this with the written description to determine issue type, severity, and the most likely responsible authority.

Review

3

The app presents a pre-filled formal complaint letter. The citizen reads, edits any details if needed, and confirms with a single tap. The photo is attached as evidence.

Route

4

The system looks up the correct government authority for the issue category and ward, then sends the formatted complaint as an email. The submission is saved to the citizen's history and pinned on the public map.

AI Processing — What Happens in Step 2

Image recognition	Identifies what is visible — pothole, garbage, broken pipe, downed wire, flooding, illegal construction, etc.
Issue classification	Assigns the issue to a category: road, water, electricity, sanitation, public safety, or environment.
Authority matching	Determines the responsible government office using the issue category + the citizen's reported ward and location.
Letter drafting	Writes a formal complaint in official English — subject line, body, date, photo evidence reference, and a clear call to action.
Severity tagging	Assigns a severity level (routine, urgent, or critical) based on safety implications in the image and description, to help prioritise routing.

Public Complaint Map

Live map view	All submitted complaints are plotted as pins on a public map, visible to any visitor without needing an account.
Category filters	Users can filter the map by issue type (road, water, electricity, etc.) to focus on what matters to them.
Upvoting	Any signed-in citizen can upvote an existing complaint to signal it affects them too, increasing its visibility and priority signal.

Complaint card	Tapping a pin shows the complaint photo, description, category, date submitted, upvote count, and current status.
Prevent duplicates	When submitting, the app checks for nearby open complaints of the same type and prompts the citizen to upvote instead of filing a duplicate.

Authority Routing — Nepal

Issue detected	Category	Routed to
Pothole / cracked road	Road	Dept. of Roads / Municipality Road Division
Water pipe break or leak	Water	Nepal Water Supply Corp (NWSC) / Ward Office
Power line fault / outage	Electricity	Nepal Electricity Authority (NEA)
Garbage pile / open sewage	Sanitation	Ward Office / Metro Solid Waste Management
Threat / crime / public disorder	Safety	Nepal Police — nearest district office
Illegal construction	Planning	Municipality / DUDBC
Environmental pollution	Environment	Dept. of Environment / Municipality

UX Considerations — General Public Focus

Simple onboarding	Sign-up in under 2 minutes — phone number + OTP or email. No lengthy forms.
Minimal steps	Complaint submission in 4 taps: photo, description, review, submit. No buried menus.
Large tap targets	Buttons and inputs sized for older users and small screens. Minimum 48px touch targets.
Clear AI output	The generated complaint letter is shown in plain language so citizens know exactly what is being sent.
Error guidance	If the AI cannot classify an image, it asks a simple follow-up question rather than failing silently.
Submission receipt	After sending, the citizen sees a clear confirmation screen with the authority name and email address used.

Feature Roadmap

Phase 1 — Launch 8 to 10 weeks • User registration + OTP login • Photo capture with GPS auto-fill • AI image + text classification • Formal complaint letter generation • Email routing to authority • Submission confirmation screen • Public complaint map with pins • Category filter on map • Duplicate detection + upvoting • Citizen complaint history page	Phase 2 — Growth 3 to 4 months • Nepali language support • Voice input via speech-to-text • Multiple photos per complaint • Offline mode + background sync • Complaint severity heat map • Anonymous submission option • Authority contact directory • Social sharing of public complaints	Phase 3 — Future Post-launch • Government API integration • Real-time status tracking • Authority response portal • Resolution rate analytics • Escalation on non-response • White-label for other countries • SMS + WhatsApp dispatch • Open data export for NGOs
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Tech Stack

Web frontend	Next.js + React, Tailwind CSS — responsive, works on low-end devices and slow connections
Mobile app	React Native + Expo — single codebase for iOS and Android
AI / Vision	Claude API — vision model for image reading, text model for classification and letter drafting
Backend	Node.js + Express REST API, PostgreSQL for data, Supabase for auth and photo storage
Map	Mapbox or OpenStreetMap — renders public complaint pins with category filters
Email dispatch	SendGrid — sends formatted complaint emails to authority inboxes
Hosting	Vercel (web + API), Cloudflare CDN, S3-compatible storage for complaint photos

Out of Scope — Current Version

• Complaint status tracking — requires government API integration (planned for Phase 3)
• Authority response portal — no government API available yet
• Nepali language support — Phase 2 feature
• SMS / WhatsApp dispatch — email only at launch

- Anonymous submission — account required for accountability

- Offline-first mode — Phase 2 feature

Final plan document — Civic Complaint & Grievance App · Nepal · Production Build · 2025. Authority names and routing logic should be verified against current Nepal government structures before deployment.